

Chapter 6 In-Class Exercise: Building a Fourier-Series Program

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1 Step 1: One Fourier Term

1. The sawtooth approximation is

$$f_N(x) = 2 \sum_{n=1}^N \frac{(-1)^{n+1}}{n} \sin(nx).$$

Begin by writing a function that evaluates only the n -th term:

```
import numpy as np
import matplotlib.pyplot as plt

def fourier_term(x, n):
    term = 2 * (-1)**(n + 1)
    term = term * np.sin(n*x) / n
    return term
```

Test the function:

```
x = np.pi / 4

print(fourier_term(x, 1))
print(fourier_term(x, 2))
print(fourier_term(x, 3))
```

- What are the two parameters of the function?
- What changes when n changes?

2 Step 2: Sum the Terms

1. Write a second function that calls `fourier_term()` repeatedly and adds the results:

```
def sawtooth(x, N):
    total = 0.0

    for n in range(1, N + 1):
        total = total + \
            fourier_term(x, n)

    return total
```

Test the function at one point:

```
x = np.pi / 4

print(sawtooth(x, 1))
print(sawtooth(x, 5))
print(sawtooth(x, 20))
```

- Which function performs one calculation?
- Which function controls the summation?
- Why does `sawtooth()` call `fourier_term()` inside a loop?

3 Step 3: Plot the Series

1. Create a third function whose job is only to plot the approximation:

```
def plot_series(x, N):
    y = sawtooth(x, N)

    plt.plot(x, y,
             label=f"N = {N}")
```

This function does not calculate the Fourier terms directly. Instead, it calls `sawtooth()`, which in turn calls `fourier_term()`.

4 Step 4: Put It Together

1. At the bottom of the program, write the main script:

```
x = np.linspace(-np.pi,
                np.pi, 400)

plot_series(x, 1)
plot_series(x, 5)
plot_series(x, 20)

plt.plot(x, x,
         label="f(x) = x")

plt.xlabel("x")
plt.ylabel("f(x)")
plt.title("Sawtooth Fourier Series")
plt.legend()
plt.grid()
plt.show()
```

- Run the complete program and compare the three approximations.
- Which function calls `fourier_term()`?
- Which function calls `sawtooth()`?
- Which part of the program controls which values of N are plotted?
- Change the final call from
`plot_series(x, 20)`
to
`plot_series(x, 50)`
and observe how the approximation changes.